

Summary	A versatile senior leader capable of delivering software systems in complex and demanding environments, experienced in every aspect of the Software Development Life Cycle. Adept at technology strategy and communication. Key Achievements include being a founding coach on the University of Cambridge Judge Business School Startup Accelerator programmes and foundational contributions to the GitOps methodology.		
Contact Information	 fractallambda.com  brice@fractallambda.com  +44 7548 312227  English, French	 linkedin.com/in/brice-fernandes/  stackoverflow.com/u/140264  github.com/bricef  @fractallambda	
Skills	LEADERSHIP & MANAGEMENT: Engineering strategy and vision, Org design and resourcing, Budget ownership, R&D leadership, Technical recruitment, Career frameworks and progression, People development and coaching, Delivery and operations, Customer Experience, Site Reliability Engineering, Incident response and business continuity, Agile principles (Scrum, Kanban and Extreme Programming practices), Lean software development, Full product development life-cycle, Technical training and skill assessment, Business modelling, Systemic innovation. GOVERNANCE & COMPLIANCE: Data protection (GDPR, ROPA, registered DPO), Security programme ownership, Cyber Essentials+, Vendor security assessment, Data residency, Risk management, Policy and audit. PLATFORMS: Cloud Native, GitOps, AWS, GCP, Multi-cloud, Hybrid Cloud, Kubernetes, Terraform, Prometheus, Grafana, Linux administration. SOFTWARE DEVELOPMENT: Functional Programming, Python, Javascript (ES6, React, NodeJS), Go, Rust, C, C#, F#, Clojure, Scheme, Test Driven Development, Git, LLM-assisted development workflows, Agent-based modelling, Natural language processing.		
Certifications	Santa Fe Institute Introduction to Agent Based Modelling Scrum Alliance Certified Agile Leadership 2 Scrum Alliance Certified Agile Leadership 1 AWS Certified Machine Learning - Specialty AWS Certified Security - Specialty AWS Certified Solutions Architect - Professional AWS Certified DevOps Engineer - Professional MIT Media Lab: AI-Generated Media Course	Completed August 2024 Valid to August 2026 Valid to August 2025 Valid to October 2025 Valid to October 2025 Valid to September 2023 Valid to October 2023 Completed October 2021	
Publications	The Evolution of Complementary Cognition: Humans Cooperatively Adapt and Evolve through a System of Collective Cognitive Search. Taylor, H., Fernandes, B., & Wraight, S. (2021). Cambridge Archaeological Journal, 1-17. doi: 10.1017/S0959774321000329		
Selected Talks	SRE Best Practices Video available	October 2020	
	Delivering Quality at Speed with GitOps Video available	August 2019	
	Building Objects With Functions An interactive version as well as a recording are available.	June 2015	

Relevant Experience

Director of Engineering at [Raspberry Pi Foundation](#)

September 2024 - Present

Engineering leader for the Foundation's Digital Products division, accountable for engineering across multiple product teams building the learning platforms used by young people and educators worldwide. Line manager to the Engineering Manager cohort, reporting to the Chief Technology and Product Officer. Owned engineering vision and strategy, org design and resourcing, the engineering budget, and the security, compliance and data-protection posture of the digital estate. Member of the foundation's leadership team, representing engineering to the executive and management board.

- Designed and landed a technology-agnostic engineering career framework (Apprentice to Principal, plus a distinct management track), now in active use across 1:1s, reviews and progression decisions
- Established a working-group operating model with charters, tiers and explicit decision rights — eight autonomous groups covering security, architecture, AI, accessibility and people development
- Owned the US data residency strategy for a public-sector opportunity of ~1M students, authoring the costed brief that framed the executive decision
- Built the Foundation-wide data map (~90 systems, 20 data categories, 25 data flows) as the evidence base for ROPA and ongoing privacy compliance
- Led the compliance and security programme for Digital Products: Cyber Essentials+, vendor security assessments, cloud access audits, software approval, business continuity and incident response
- Directed engineering AI adoption through an adoption plan, a dedicated working group, and an adoption-maturity reporting tool built with external partners
- Grew and stabilised the team through recruitment at every level, contractor procurement, and structured investment in people development, with measurable gains in retention

KEYWORDS: *Engineering Leadership, Org Design, Career Frameworks, People Development, Recruitment, Engineering Strategy, Architecture, Data Protection, GDPR, ROPA, Compliance, Cyber Essentials+, Security, Budget Ownership, Delivery Management, AI Adoption, Education, Non-profit*

DevOps Practice Leader at [Capgemini](#)

January 2022 - June 2024

Senior Leader of the DevOps practice for Capgemini UK Digital Customer Experience (DCX). Worked on commercial proposition development, aligning product, service and capability with both clients, market units, and other delivery areas in DCX and beyond. Owned the technology radar and external thought leadership for DevOps practice within the DCX delivery unit. Delivered client engagements for some of the UK's largest organisations.

- Worked on all aspects of capability management (Products and offers, culture, training, recruitment, career framework)
- Senior technical leader on client engagements
- Owned technical architecture and ways of working on client engagements
- Security cleared to work on National Critical Infrastructure
- Managed and mentored other senior leaders
- Became Capgemini Global DevOps Community's UK point of contact

KEYWORDS: *Senior Leadership, Capability Management, Architecture, Delivery Management, Digital Transformation, Recruitment, Product Management, National Critical Infrastructure*

Senior Solution Architect at [Weaveworks](#)

September 2018 - September 2021

Architected, developed and delivered hybrid cloud application platforms for some of the company's most high-profile clients, including in multiple regulated industries. Responsible for both project technical leadership and client relationships.

Worked on all aspects of commercial engagements, from pre-sales to delivery management. Was responsible for the design and delivery of GitOps and Cloud Native training for clients and partners.

- Provided architecture consultancy for internal application platforms to global banks
- Led the design and architecture of the internal application platform to support the autonomous vehicle project of one of the world's largest companies
- Led the initial design and architecture workshop for the re-platforming effort of a large European telecommunication provider
- Defined the principles of GitOps
- Trusted key-holder for critical internal systems
- Designed and implemented customer experience playbook
- Designed and delivered public and private Cloud Native training

KEYWORDS: *GitOps, Customer Experience, Delivery management, Kubernetes, Cloud Native, Pre-sales engineering, Monitoring, Observability, Digital transformation, SRE, Training, Recruitment*

Senior Development Engineer at [Weaveworks](#) September 2017 - September 2018
Developed Go tools for managing Kubernetes. Helped define the GitOps methodology and communicate it to the developer community. Presented GitOps to conferences in the UK and internationally and delivered training on Kubernetes and GitOps. Member of the on-call engineering team directly responsible for the uptime and maintenance of our online SaaS platform.

KEYWORDS: *Kubernetes, GitOps, Cloud Native, Prometheus, Terraform, Go, On-Call, Incident response, DevOps, SRE, Developer Experience, Training, Recruitment*

Founder at Decacoder October 2015 - January 2017
Founded an online education company delivering high quality online education to developers via a custom platform. I led the development of a scalable media delivery platform from scratch while also being responsible for all other business areas. I also set up a continuous integration and continuous deployment pipeline using AWS services, and was responsible for the educational design and user experience of all students using the platform. Also became registered Data Protection Officer for the company.

KEYWORDS: *Product development, Leadership, Javascript, ES6, React, Babel, Firebase, Continuous integration, Continuous deployment, AWS, Gulp, Stripe, Education, UX*

Startup Mentor & Founding Coach, [Accelerate Cambridge](#) 2013 - 2016
I was selected to be a founding coach and mentor for the University of Cambridge Judge Business School startup accelerator Accelerate Cambridge. I delivered a startup founder education curriculum and coached teams on product design, technology strategy, development process, team management, pitching for investment and clear communication. Several of the startups I mentored received prestigious awards, including winning the Pitch@Palace competition.

KEYWORDS: *Startups, Coaching, Business modelling, Pitching, Leadership, Investment, Technology Strategy*

Instructor & Coach, [Ignite Programme](#) July 2015, July 2017
I was an instructor and coach for the University of Cambridge Judge Business School Ignite programme. The Ignite programme is a one week course aimed at transforming existing early stage enterprises through high quality lectures and challenging workshops. I was responsible for leading workshops during the programme, ensuring that each attendee adopted the knowledge shared into their business practice.

KEYWORDS: *Startups, Education, Coaching, Leadership, Event facilitation, Business Modelling*

Founder, [Cambridge Programmer Study Group](#) July 2014 - December 2016
Founded the most active programming meetup in Cambridge, meeting twice a week to study computer science topics. Over two years we studied Machine Learning, Cryptography, Concurrency and Functional programming, amongst other topics.

KEYWORDS: *Machine Learning, Functional Programming, Scheme, SICP, Neural Networks, Cryptography, Concurrency, Computer architecture, Compilers, Interpreters, Community building*

Instructor, F# developer course September 2015 - August 2016
Successfully kickstarted an online tutorial series on the F# programming language. De-

signed and delivered a curriculum that taught the real world applications the F# language and functional programming, including type-driven design.

KEYWORDS: *F#, Functional Programming, Education, Screencasting, Kickstarter, Video production*

Instructor, Unity3D Development Course

August 2014 - August 2015

Successfully kickstarted an online tutorial series on game development using the Unity3D game engine. Designed a curriculum to teach programming for game development in the C# language, including the fundamentals of computer science to a novice audience. The project was delivered over the Udemmy.com online platform and reached over 100 000 students. The complete course was accredited for higher education credits and worth 10 European Credit Transfer System credits (~10% of an undergraduate degree).

KEYWORDS: *Education, Unity3D, C#, Game Development, Screencasting, Kickstarter, Video production*

Software Engineer, Cambridge Broadband

March 2011 - October 2013

Developed and maintained a wide range of software to enable and support a microwave backhaul product at Cambridge Broadband. Worked on cross platform graphical user applications, embedded C network management using SNMP, internal tools and existing product maintenance. Was also part of a process improvement team that re-designed the software and product development process for the company. Took responsibility for technical recruitment of C developers, drafting requests for quotes and selecting and managing contractors.

KEYWORDS: *C, PPC, Embedded, Python, Linux, GTK, SNMP, Process improvement, Agile, RFQ, Recruitment*

Education

**The University Of Sheffield
BSc. Physics with Computer Science**

2006 - 2010